

JEE MAIN REVISION MASTER: CARBOXYLIC ACIDS AND ITS DERIVATIVES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Acidity of Carboxylic Acids & Substituent Effects	6 PYQs	Acid-Base Properties
Moderate	Reactions of Carboxylic Acids & Derivatives	6 PYQs	Syntheses & Reactions

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Acidity of Carboxylic Acids & Substituent Effects	$\text{Acidity: } \text{HCOOH} > \text{CH}_3\text{COOH} > \text{CH}_3\text{CH}_2\text{COOH}$
Reactions of Carboxylic Acids & Derivatives	$\text{R-CH}_2\text{-COOH} \xrightarrow{\text{Br}_2 / \text{Red P}} \text{R-CH(Br)-COOH} \quad (\text{HVZ Reaction})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Acidity of Carboxylic Acids & Substituent Effects (Moderate Yield — 6 PYQs)

Core Principle: Carboxylic acids are much more acidic than phenols because the negative charge in carboxylate ion is delocalized over two highly electronegative oxygen atoms. Electron withdrawing groups (-I, -M) increase acidity; electron donating groups (+I, +M) decrease acidity.

■ **Caution / JEE Pitfall:** Formic acid (HCOOH) is more acidic than acetic acid (CH_3COOH) because the methyl group exerts an acid-weakening +I effect.

■ **Speed Trick:** Carboxylic acids effervesce with NaHCO_3 (liberating CO_2 gas), whereas phenols (except picric acid) do not.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Carboxylic acids are more acidic than phenols because..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following has most acidic Hydrogen ?..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "The reagent which can do the conversion $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{-CH}_2\text{-OH}$ is..." [Ans: N/A]

3.2 Reactions of Carboxylic Acids & Derivatives (Moderate Yield — 6 PYQs)

Core Principle: HVZ reaction: α -halogenation of carboxylic acids with X_2 / Red P. Decarboxylation with soda lime (extNaOH + CaO, 3:1) yields alkane with one fewer carbon. Reduction of esters with DIBAL-H at low temperature gives aldehydes selectively.

■ **Caution / JEE Pitfall:** Benzoic acid and formic acid cannot undergo HVZ reaction because they possess no α -hydrogens.

■ **Speed Trick:** DIBAL-H reduces esters and nitriles to aldehydes, not to alcohols or amines.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Statement - I : Reduction of ester by DIABL-H followed by hydrolysis gives aldehyde. Statement - II : Oxidation of benzyl alcohol with aqueo..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "Ethanoic acid undergoes Hell-Volhard Zelinsky reaction but methanoic acid does not, because of..." [Ans: N/A]
- [2022 Shift] Q#4 (MCQ): "Reaction by which benzaldehyde cannot be prepared is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Carboxylic acids are more acidic than phenols because

- (A) Formation of dimers
- (B) Intermolecular hydrogen bonding
- (C) More covalent nature
- (D) More resonance stabilisation of their conjugate base

Correct Answer: None

Solution Analysis: Carboxylic acids are more acidic than phenols because their conjugate base is more resonance stabilised. So the correct option is: Option D: More resonance stabilisation of their conjugate base When a carboxylic acid loses a proton, it forms a carboxylate ion: $RCOOH \rightarrow RCOO^- + H^+$ In the carboxylate ion, the negative charge is delocalised equally over two oxygen atoms: $R-C(=O)-O^- \leftrightarrow R-C(O^-)=O$ Both resonance structures are equivalent, so the carboxylate ion is highly stable. Phenol also forms a conjugate base, called phenoxide ion:

Benchmark Problem 2: KCET 2025 (Q#2)

Statement - I : Reduction of ester by DIABL-H followed by hydrolysis gives aldehyde. Statement - II : Oxidation of benzyl alcohol with aqueous $KMnO_4$ leads to the formation to Benzaldehyde. Among the above statements, identify the correct statement.

- (A) Both statements - I and II are false
- (B) Statement - I is true but statement - II is false
- (C) Statement - I is false but statement - II is true
- (D) Both statements - I and II are true.

Correct Answer: None

Solution Analysis: Statement I is true, and Statement II is false. DIBAL-H (diisobutylaluminium hydride) at low temperature selectively reduces an ester to the corresponding aldehyde upon work-up. Aqueous $KMnO_4$ oxidizes benzyl alcohol all the way to benzoic acid, not benzaldehyde. Answer: Option B.